

Bachelor Thesis

April 7, 2026

From Security by Obscurity to Imminent Exposure:

Security Risks for IoT Devices in the IPv6
Transition

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URL: <https://github.com/Vite04/ThesisCode>

Project period: 2026-01-15 - 2026-07-01

Security and Privacy of Information, Networks, and Systems (SPINS)

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Acknowledgements

Abstract

Contents

1	Introduction	1
1.1	The IPv4 to IPv6 Transition	1
1.2	Loss of NAT and Increased IoT Exposure	1
1.3	Research Objectives	1
1.4	Contributions	1
1.5	Thesis Structure	2
2	Background and related work	3
2.1	The Loss of Perimeter Security	3
2.2	The IoT Security Landscape	3
2.3	IPv6 Visibility & Identifiability	4
2.4	Scanning & Target Generation in IPv6	5
2.5	Synthesis & Research Gap	6
3	Methodology	7
3.1	Research Design	7
3.2	Testbed Architecture and Threat Model	7
3.3	Experimental Scenarios	8
3.4	Measurement and Evaluation	10
3.5	Ethical Considerations	11
4	Experimental Results	13
4.1	Address Assignment and Identifier Exposure	13
4.2	Internal Baseline and Local Service Availability (S0)	13
4.3	External IPv6 Reachability with Firewall Enabled (S1)	14
4.4	External IPv6 Accessibility with Firewall Disabled (S2)	14
4.5	Local Traffic Visibility and Plaintext Exposure (S3)	15
4.6	IPv4 Baseline Without Port Forwarding (S4)	15
4.7	Cross-Scenario Summary	15
5	Discussion	17
5.1	Interpretation of Results	17
5.2	Threats to Validity	17
6	Limitations and Mitigation	19
6.1	Limitations	19
6.2	Recommendations for Network Administrators	19
6.3	Recommendations for Manufacturers	19
6.4	End-User Best Practices	19
7	Conclusion	21
7.1	Future Work	21
	Bibliography	23

A	Source Code	25
B	Experimental Results	35
B.1	Scenario 0: Baseline (Internal Network)	35
B.1.1	Local Network Discovery and ICMPv6	35
B.1.2	Internal Service Verification	39
B.2	Scenario 1: External Reachability (Firewall ON)	40
B.2.1	Remote Reachability and Service Probing	40
B.3	Scenario 2: Service Exposure (Firewall OFF)	41
B.3.1	External Service Access	41
B.3.2	Packet Analysis and Remote Discovery	42
B.4	Scenario 3: Local Traffic Observation	42
B.4.1	Deep Packet Inspection (DPI)	42
B.5	Scenario 4: Comparative Baseline (IPv4 / NAT)	57
B.5.1	IPv4 External Reachability Attempt	57

1. Introduction

This thesis explores how global IPv6 addressing increases the attack surface of the Internet of Things (IoT) environment. The study combines a comprehensive literature review and an empirical assessment of default system configurations through controlled laboratory demonstrations, highlighting how the adoption of new network standards can potentially affect the security of millions of end IoT devices.

1.1 The IPv4 to IPv6 Transition is the result of an inherent limitation in the design of IPv4's address space, which only supports 2^{32} unique addresses. A boundary already exceeded by the estimated 16 billion IoT devices alone in 2024 [1]. IPv6 mitigates this limitation by implementing a 128-bit architecture, offering a virtually infinite address space. Despite its technical advantage, the global adoption has been gradual due to the current scale of IPv4-based infrastructures and the cost of migration. Therefore, technologies such as Network Address Translation (NAT) were adopted to extend the lifetime of IPv4 [2].

In practice, NAT allows multiple internal hosts within a private local-area network (LAN) to share a single public IPv4 address by translating internal IP addresses and TCP/UDP port numbers to a public address and port at the network boundary. This translation process is stateful and connection-oriented, meaning that the address mappings are created only for outbound traffic initiated by hosts within the LAN [2], [3]. Consequently, unless explicitly allowed through static port-forwarding rules, any unsolicited inbound traffic that does not match an existing mapping entry is discarded by default [3]. Nonetheless, the use of NAT translation tables violates the original end-to-end connectivity principle of the Internet architecture [4], and simultaneously provides a form of perimeter protection by preventing direct external access to internal hosts.

1.2 Loss of NAT and Increased IoT Exposure reflects a shift in network architecture paradigms. Until now, many IoT devices have been designed under an implicit threat model that assumes limited inbound reachability, relying primarily on network topology, firewall configurations, and traffic routing rather than on end-device security mechanisms. With the deprecation of NAT in IPv6, this assumption no longer holds. Depending on the network configuration, devices become directly reachable without any change to their internal security properties, thereby increasing their effective exposure [4], [5], [6]. The direct exposure of IoT devices is especially problematic due to the attack vectors commonly associated with this ecosystem. A large number of IoT devices have exposed services that are only protected by weak or default authentication mechanisms, enabling large-scale automated exploitation [7]. Embedded devices often contain insecure network services or misconfigurations, as well as persistent vulnerabilities produced by design constraints and limited update mechanisms [8]. Poorly secured management interfaces and API calls significantly expand the attack surface when IoT devices become externally reachable [9]. It is precisely the interaction between inherent device vulnerabilities and expanded reachability that increases the risk of remote exploitation.

1.3 Research Objectives The main objective of this thesis is to analyze how the transition to globally routable IPv6 addressing increases the attack surface of IoT devices by enabling direct external exposure in real-world network environments. Specifically, this study aims to identify under which router configurations, network parameters, and default device settings IoT devices become directly accessible from external networks.

1.4 Contributions TODO

1.5 Thesis Structure Chapter 2 provides the necessary technical background on IPv4, IPv6, modems configurations and IoT security models. Chapter 3 describes the experimental methodology and demo setup used to evaluate device exposure under different network configurations. Chapter 4 presents the empirical results and analyzes the conditions under which IoT devices become externally reachable. Chapter 5 discusses the security implications of these findings and outlines practical considerations for reducing unintended exposure. Finally, Chapter 6 concludes the thesis and summarizes key insights and possible future developments for the research.

2. Background and related work

2.1 The Loss of Perimeter Security The security of modern networks has been built around a well-defined perimeter, a topological barrier that separates trusted local networks (LAN) from wider, potentially hostile networks (WAN). However, the transition toward IPv6 alters this defensive boundary by removing legacy mechanisms and creating critical asymmetries in firewall policies. For decades, NAT protected IoT devices by defaulting to deny inbound traffic. However, with IPv6's direct global addressing, this implicit security disappears. As the research below demonstrates, organizations have failed to compensate with explicit firewall policies, creating a "perfect storm" of exposure [6], [10].

The elimination of NAT creates the need for IPv6 networks to implement filtering policies and firewalls that are equivalent to, or stronger than, those used in IPv4. Nonetheless, empirical research shows a substantial gap in the enforcement of these organizational policies. A large-scale characterization of dual-stack networks analyzing the behavior of 520,000 servers and 25,000 routers demonstrated that corporate security policies in IPv6 are consistently more permissive [10]. In particular, several high-value target applications were identified with more permissive security policies in IPv6. Management protocols commonly targeted by attackers and botnets, such as SSH, Telnet, and SNMP, were found to be more than twice as exposed on routers in IPv6 as they are in IPv4 [10]. This systemic weakness suggests that organizations are more inclined to prioritize securing legacy infrastructure while leaving IPv6 deployments insufficiently protected due to permissive configurations or technical and operational oversight [10].

While corporate networks already show significant security gaps, the situation in residential networks is more concerning. Recent studies on residential IPv6 deployments highlight significant security risks related with the residential transitions. Differently from IPv4 NAT, which by default imposes a default-deny access control model, IPv6 residential router manufacturers (CPE) have the discretion to decide whether to implement restrictive or permissive filtering policies. After actively scanning more than 2.5 million residential IPv6 prefixes, the researchers found that many routers adopted a default-allow policy [6], deliberately routing unsolicited incoming traffic from the WAN to the LAN. The study further showed that simple IPv6 scans allowed researchers to directly reach end devices such as printers, smart light bulbs, and CCTV systems, surpassing the discovery rates that previously required exhaustive scanning of the entire IPv4 address space. These findings indicate that the residential transition weakens traditional firewall protections, exposing additional hosts to botnets and increasing overall Internet security risks [6].

2.2 The IoT Security Landscape The loss of the network perimeter described above conflicts with the current state of the embedded device ecosystem (IoT). The design of these devices often assumes that their operating environment will remain isolated and private, resulting in firmware architectures and communication protocols that are vulnerable when directly exposed to the Internet through IPv6.

Security practices in home IoT devices have been analyzed by Alrawi et al. [11]. After evaluating 45 commercial devices across four components: the device itself, the mobile application, the cloud endpoint, and the communication channel, the authors found that manufacturers often assume that the LAN environment is inherently trustworthy and protected by a boundary router. As a result, many devices rely primarily on the domestic router as the main security mechanism. This assumption appears as a consequence of prioritizing product release time over security considerations, which leads to the omission of internal security controls. Many devices lack local encryption mechanisms, making them susceptible to man-in-the-middle attacks, while mobile applications frequently request excessive privileges that go beyond their intended functionality [11]. These results are not limited to unbranded IoT devices. Fernandes et al. also showed that platforms such as Samsung SmartThings have inherent design flaws in their permission model that result in over-privileged third-party applications, allowing potential remote exploitation when the devices are

externally accessible [9]. Considering these findings together with the disappearance of perimeter filtering policies, the implication is that devices and platforms designed under the assumption of network isolation become directly exposed to remote attackers.

In the application layer, the technical debt of the embedded ecosystem increases the viability of exploitation. Hardware limitations and accelerated development cycles lead to devices running obsolete or misconfigured network services. Costin et al. examined 32,000 firmware images and discovered 38 previously unknown vulnerabilities in more than 693 firmwares, confirming that latent vulnerabilities persist due to outdated software and the fact that security is systematically sacrificed for cost and time-to-market [8]. The results are consistent with the deficiencies found by Alrawi et al. [11], whose evaluation of 45 devices revealed a persistent reliance on plaintext protocols such as HTTP instead of implementing TLS encryption, as well as the use of legacy administrative services such as Telnet. In traditional IPv4 networks behind NAT, interacting with these vulnerabilities would require prior access to the LAN. However, in the context of globally routable IPv6 networks, these insecure services can be directly exposed, facilitating credential extraction and the propagation of automated malware. The Mirai botnet demonstrated the consequences of this type of exposure, using a dictionary of only 62 pairs of default credentials for brute-force attacks via Telnet, the malware was able to infect more than 600,000 IoT devices and execute massive DDoS attacks against high-profile targets [7].

Even when manufacturers protect public web interfaces with passwords, devices may still expose residual risk through hidden administrative channels, as documented by the EAGLEYE study [12]. Hidden network interfaces are defined as functional access routes to the device that are neither documented nor accessible through the standard navigation of the main portal, often introduced during development but later abandoned in production [12]. Because they are not exposed through the usual interaction flow, these interfaces evade standard authentication controls. EAGLEYE analyzed 13 commercial devices and discovered 79 hidden interfaces, 25 times more than previous tools commonly used. The activation of these hidden channels triggered 29 zero-day vulnerabilities, including command injection, XSS, information leakage, and operating backdoors that provided full access to the device, resulting in seven assigned CVEs [12]. The presence of this type of authentication-bypass vulnerability in embedded firmware was previously validated by Shoshitaishvili et al. using Firmalice, a binary analysis framework that, when evaluating commercial devices, detected backdoors that allowed administrative access through specially crafted packets without requiring legitimate credentials [13]. When these findings are considered together, the erosion of the local network perimeter not only exposes known vulnerabilities in IoT devices but also opens the path for potential system takeover and credential bypass.

2.3 IPv6 Visibility & Identifiability Even considering the scenario described above, a defensive argument that justified several of the shortcomings of the IoT ecosystem was that, regardless of its vulnerabilities, the 128-bit IPv6 address space provided a form of practical invisibility, in a role similar to NAT in IPv4, different in mechanism but often assumed to produce a comparable protective effect [14]. Nonetheless, core properties of IPv6, particularly its reliance on Stateless Address Autoconfiguration (SLAAC) without mandatory cryptographic authentication [5], [15] and the deterministic embedding of hardware MAC addresses via EUI-64 [16], make devices highly identifiable and enable the construction of precise visual and topological profiles for attackers.

In order to guarantee plug-and-play functionality, IPv6 [5] relies on Stateless Address Autoconfiguration (SLAAC) [17], a mechanism that uses mandatory messages in the Neighbor Discovery Protocol (NDP) [18], specifically Router Solicitations (RS) and Router Advertisements (RA). The structural weakness lies in the fact that the IPv6 standard does not require cryptographic authentication for these local network exchanges. The threats resulting from this absence were formally cataloged by the IETF [15], which documented how non-privileged actors within the same network segment can orchestrate man-in-the-middle attacks through falsified NDP messages, sending fake RA packets to reroute traffic or forcing denial of service during the autoconfiguration process. Even though Secure Neighbor Discovery (SEND) [19] was proposed as a mitigation mechanism, its implementation complexity and its architectural requirement for public keys

have limited its practical deployment. The issue is further complicated because indiscriminately blocking ICMPv6 [20] would break network connectivity, since NDP depends on these messages to operate [21]. As a result, the network initialization mechanisms used by IPv6, and therefore by many IoT devices, represent a permanently exposed attack surface.

The SLAAC process requires that the device generates its own interface identifier (IID) to use the last 64 bits of the IPv6 address. Traditionally, the industry has implemented the EUI-64 standard, which directly embeds the physical address (MAC) of the hardware into the public IID. The privacy and security implications of this design were discussed in the study *IPv6SeeYou* [16]. The researchers demonstrated that this mechanism produces a systematic leakage of MAC addresses, allowing an unprivileged remote adversary to geolocate residential networks and IoT hosts with street-level precision, identifying more than 12 million routers in 146 countries with an accuracy of within 39 meters [16]. By correlating the Organizationally Unique Identifier (OUI) with IPv6 geolocation databases and Wi-Fi positioning data, the investigation showed that an identifier at the network layer had become a direct vector for physical tracking and the identification of the device manufacturer. Once these identification capabilities are combined with firmware vulnerabilities (Section 2.2), a remote attacker could not only discover the IoT device but also determine the most likely manufacturer, firmware, and physical location, enabling highly targeted attacks.

The visibility of the connected ecosystem is further increased by the traffic generated at the application layer that modern protocols require to function properly. IoT devices use multicast protocols such as mDNS and SSDP for service discovery within the local network [22]. Yu et al. demonstrated how this passive traffic forms a network fingerprint [22]. Through the analysis of structural and textual information embedded in broadcast and multicast packets, their proposed framework was able to identify with high accuracy the brand, model, and category of IoT and mobile devices without actively interacting with them [22]. In an IPv6 environment where devices are directly reachable, this passive profiling capability complements manufacturer identification through EUI-64 [16], giving an attacker a detailed map of the device ecosystem before launching any exploit.

Summarizing, these findings destroy the illusion of opacity. In an IPv6 environment, an IoT device not only receives a publicly reachable address by default, but its autoconfiguration mechanisms also expose the identity of its manufacturer through EUI-64 addresses [16]. At the same time, NDP exchanges do not require mandatory cryptographic authentication [15], and broadcast protocols can passively reveal the device's brand, model, and category to any listener on the network segment [22]. As a result, the assumed "invisibility" of IoT devices disappears by design in the IPv6 environment.

2.4 Scanning & Target Generation in IPv6 The scanning and discovery phase is the first step in the chain of a cyberattack. Under IPv4, with an address space of only 2^{32} addresses, tools such as ZMap showed that it was possible to scan the entire Internet in less than one hour [23]. IPv6, with its much larger address space, was designed under the assumption that its virtually unlimited size would make exhaustive enumeration computationally impractical, turning address space size into a form of implicit security. However, recent studies have shown that this mathematical barrier has been systematically overcome through smart and adaptive scanning techniques.

The capability to scan the entire IPv4 space expanded large-scale Internet measurement. Building on tools such as ZMap [23], platforms like Censys allowed the indexing of vulnerabilities, certificates, and embedded devices across the Internet [24]. This visibility over the IPv4 address space made it possible to identify services and devices exposed at a global scale. Botnets such as Mirai exploited the same principle, locating IoT devices such as IP cameras by performing simple stochastic scans and compromising them through default Telnet credentials, as discussed in Section 2.2.

The IPv6 address space was designed under the idea that it would be too large to make exhaustive mapping computationally feasible. Nevertheless, recent research has shown that this space can still be mapped through adaptive scanning systems. One of the most notable contributions is *6Sense* [14]. One of the main challenges in IPv6 scanning is aliasing, where intermediate routers that are poorly configured respond to scans directed at tens of millions of inactive addresses within a prefix, diluting the resources of the scan-

ner with false positives [14]. 6Sense addresses this problem through a reinforcement learning approach combined with online and offline de-aliasing modules that can distinguish between illusory prefixes [14]. By dynamically learning active addresses and focusing on predictable assignment patterns—such as lower-order bytes used by IoT interfaces or grouping patterns in nybbles—the system was able to discover millions of active IPv6 hosts, approximately 3.6 times more than previous methods, and successfully correlate them with critical vulnerabilities (CVEs) [14]. Furthermore, Pan et al. demonstrated that side channels derived from ICMP rate limiting in routers allow the structure of IPv6 networks to be inferred without the need for direct scans [25]. These results establish that the virtually unlimited IPv6 address space is not a viable mechanism for obscurity.

2.5 Synthesis & Research Gap The analysis of the literature and related work presented in the previous sections reveals potential issues associated with the transition to IPv6. The previous sections examined four dimensions of this problem individually: the disappearance of the network perimeter within which the IoT environment was originally developed (Section 2.1), the inherent vulnerabilities of IoT devices (Section 2.2), the identifiability of devices in IPv6 networks (Section 2.3), and the feasibility of large-scale scanning in the IPv6 address space (Section 2.4). When considered in isolation, each of these phenomena represents a considerable risk. However, when examined together, a qualitatively different picture emerges. Despite this body of work, a gap in current research methodologies remains.

In a complete IPv6 environment, an IoT device connected to a residential network automatically receives a globally routable address, and in many cases the router does not filter incoming traffic [6], [21]. At the moment the device is autoconfigured through SLAAC, which relies on mechanisms that lack cryptographic authentication [15], the identity of the manufacturer may already be exposed through EUI-64 [16], while its discovery protocols may have already passively revealed its brand, model, and category [22]. At the same time, specialized scanning tools such as 6Sense can locate active devices within the 128-bit IPv6 address space [14]. Once discovered and identified, the device presents the same vulnerabilities extensively documented in the literature: firmware that runs unsafe protocols or lacks encryption [11], default credentials that can be exploited to obtain administrator privileges as demonstrated by the Mirai botnet [7], hidden interfaces that bypass authentication [12], and platforms with deficient permission hierarchies [9].

Each link of this chain has been individually validated, but the full chain remains unexplored. Network-focused studies concentrate on scanning feasibility and filtering policies without evaluating what happens once a device is reached. Device-focused research analyzes firmware vulnerabilities assuming an isolated IPv4 local network. No existing study examines, under controlled conditions, which specific router configurations and device settings enable or prevent the external reachability of an IoT device in a residential IPv6 environment.

This thesis addresses this gap through a controlled laboratory experiment using commercial hardware (ESP32), simulating being an IoT device, connected to real residential IPv6 infrastructure. By systematically varying router configurations and device parameters, this work identifies the specific conditions under which an IoT device with a globally routable IPv6 address becomes externally accessible, and demonstrates the security implications of that exposure.

3. Methodology

3.1 Research Design This chapter describes the methodology used to empirically evaluate the external reachability and service exposure of an IoT device in a residential IPv6 network configuration. The study follows a controlled case study design, in which a programmable embedded device (ESP32) is used as a representative IoT system. As discussed in Section 2.2, such devices commonly exhibit simplified network stacks and weak service configurations, making them suitable proxies for real-world IoT deployments. The device is systematically configured under different network conditions, while measuring its observability, accessibility, and network traffic visibility in each case. These configurations modify key parameters such as firewall state, access origin, and interaction protocols. The experimental setup is implemented in a physical laboratory environment using commercial hardware, including a residential router and an external VPS. This approach allows precise isolation of network variables while maintaining environmental control and reproducibility. At the same time, it avoids ethical concerns associated with scanning or interacting with third-party devices on the public Internet. The experimental design is structured around a set of progressive scenarios, ranging from an internal baseline to externally reachable IPv6 configurations and a final comparative scenario under IPv4. Each scenario is evaluated based on observable network and service interactions, including reachability at the ICMP level, TCP connectivity, and the presence of application-layer responses, allowing consistent expected results across repeated executions. All evidence is collected in raw form, including terminal outputs, packet captures, and device-level logs.

3.2 Testbed Architecture and Threat Model The testbed consists of four main components: an ESP32-based device, a residential router acting as the perimeter network controller, an internal host connected to the same LAN, and an external VPS located outside the residential network. This architecture allows the behavior of the embedded device to be observed from two complementary perspectives. The first was a local perspective, which made it possible to verify the internal availability of the IoT services and inspect traffic within the LAN, and, when communication between the embedded device and the internal host was not encrypted, to examine the content of the exchanged packets. The second was a remote perspective, representing potential access from outside the residential network.

Table 3.1: Testbed Components

Component	Role	Network Position	Notes
ESP32	Target IoT device	Residential WiFi; acquires global and link-local IPv6	Runs custom firmware; exposes HTTP (80) and Telnet (23)
Residential router	Perimeter controller	Provides native IPv6 and public static IPv4	Configurable IPv6 firewall (key independent variable)
Internal host	Local observer	Same LAN as the ESP32	Used for local validation and traffic inspection
External VPS	External host	Outside residential network; native public IPv6	Used for external interactions and IPv4 comparison

The target device was an ESP32 microcontroller executing custom firmware developed for this study.

Although the implementation is custom, its design follows the IoT service models and practices described in Section 2.2, rather than introducing new mechanisms. The firmware is implemented using the Arduino environment and the ESP32 network stack, including `WiFi.h`, `esp_event.h`, `esp_netif.h`, and the lwIP socket interface. These libraries enabled the management of the WiFi connection, network events, IPv6 initialization, and the creation of TCP services over IPv6 sockets. Once connected to the wireless network, the device automatically obtains both a link-local address and a global IPv6 address, and logs this information together with its MAC address through the serial output. This allows direct documentation of the device's network identity and addressing state within the experimental setup. At the application layer, the firmware implements two services over IPv6. The first is an HTTP server running on port 80, accessible through the main route `/`, which returns a simple dashboard with device information. Furthermore, the same server exposes an undocumented endpoint, `/hidden_admin`, reflecting a pattern commonly observed in IoT devices [12]. The second service is a Telnet-like TCP server on port 23, which requests credentials in plaintext and uses default credentials (`admin/admin`). Both services record inbound connections, including the source IPv6 address and remote port, as well as relevant events such as access to the hidden endpoint, authentication attempts, and executed commands. This enables the correlation between what is externally observable at the network level and the evidence generated locally by the device itself. The residential router is the perimeter component of the testbed. This device provides native IPv6 connectivity and a public static IPv4 address, which is essential for establishing a fair comparison with the IPv4 environment. In the case of IPv6, the router allows the firewall state to be modified explicitly, making this configuration one of the key independent variables of the experiment. In the case of IPv4, incoming connections are not directly possible by default unless an explicit port forwarding rule is configured. Since no such rule was enabled in this experiment, the IPv4 setup provides a useful baseline for contrasting both exposure models without introducing additional configuration elements that fall outside the scope of this study. Within this established perimeter, an internal host is also present. This host serves a double function. On the one hand, it is used to establish the local baseline of the experiment, meaning that it verifies that the device responds correctly at both the network and application layers before any external reachability test is performed. On the other hand, depending on the scenario, it acts as a passive observer, allowing traffic within the LAN to be captured and analyzed in order to verify whether the information exchanged via HTTP and Telnet remains visible in plaintext. The external VPS, located in a different country and equipped with native IPv6 connectivity, serves as the remote observation point and is used to generate ICMPv6 probes, TCP connection attempts, HTTP requests, and Telnet sessions from outside the residential network. From this architecture, two complementary observation modes are defined. The first corresponds to an active external observer, represented by the VPS, which attempts to reach the device through its global IPv6 address and evaluate its reachability, TCP connectivity, and response at the application layer. This same external position is also used in the final IPv4 comparison. The second corresponds to a passive local observer within the LAN. This distinction makes it possible not only to evaluate to what extent the device can be reached from outside the residential network, but also whether its communications remain visible inside the local network in the absence of encryption.

3.3 Experimental Scenarios The experiment is organized into five progressive scenarios (S0 to S4). Instead of evaluating each network parameter individually, the independent variables, such as the IPv6 firewall state, the origin of access, and the protocol used, are manipulated through these scenarios. In this way, each scenario represents a controlled condition under which changes in device reachability, service visibility, and traffic exposure can be observed.

Table 3.2: Experimental Scenarios

Scenario	Perimeter Condition	Access Origin	IP Context	Objective
S0 — Internal Baseline	not relevant	internal host	IPv6	confirm device identity, global IPv6 acquisition, and local availability of the implemented services
S1 — External Firewall ON	firewall enabled	external VPS	IPv6	evaluate whether the device remains observable despite TCP service filtering
S2 — External Firewall OFF	firewall disabled	external VPS	IPv6	evaluate whether the same global IPv6 address enables effective access to HTTP, Telnet, and the hidden endpoint
S3 — Passive LAN	not relevant	internal passive observer	IPv6	determine whether HTTP and Telnet communications remain visible in plaintext inside the LAN
S4 — IPv4 Baseline	no port forwarding	external VPS	IPv4	establish a contrast with residential IPv4 by evaluating whether equivalent direct inbound access exists

Scenario 0 establishes the internal baseline for the experiment. In this phase, the device is observed only within the LAN, with the objective of verifying that IPv6 connectivity is correctly initialized, that a global address has been acquired, and that the services implemented in the firmware respond as expected at both the network and application layers. This scenario allows us to ensure that any differences observed in subsequent phases are not due to device failures or misconfiguration, but rather the result of changes in the independent variables. Scenario 1 evaluates the behavior of the embedded device in the presence of an active external observer when the IPv6 firewall remains enabled. Under this configuration, the remote VPS attempts to reach the global address of the device through ICMPv6, TCP connections, and application-layer requests via the implemented HTTP and Telnet services. The objective of this scenario is to determine to what extent the device remains observable from outside the residential network, and what information can still be obtained despite the presence of firewall filtering. Scenario 2 maintains the same external origin and the same set of interactions, but with the IPv6 firewall disabled. This allows isolating the effect of changes in perimeter policies without modifying the firmware, the device, or the location of the observer. In this phase, it is evaluated whether the global address enables effective access to the HTTP and Telnet services, as well as to the hidden interface exposed through the endpoint `/hidden_admin`. Scenario 3 changes the observation perspective to a local context and uses an internal host as a passive observer. Differently from the previous scenarios, the goal here is not to measure remote accessibility, but rather traffic visibility. This scenario allows verifying whether the information exchanged with the device, including

HTTP requests, server responses, Telnet credentials, and executed commands, remains visible in plaintext when no encrypted communication is used, as commonly observed in legacy IoT services such as Telnet [7]. The final scenario (S4) introduces a reference condition under IPv4. In this phase, the same external VPS attempts to access the device through a public and static IPv4 address, while maintaining the absence of port forwarding rules. The objective of this scenario is to evaluate whether the exposed services can be reached under typical residential IPv4 conditions, or whether they remain effectively isolated due to Network Address Translation (NAT), which restricts unsolicited inbound connections unless explicit port forwarding is configured.

3.4 Measurement and Evaluation The evaluation of each scenario will be done through three complementary levels: reachability at the network layer, the accessibility of the services, and the traffic visibility at the application layer. This distinction is important because it reflects different stages of exposure within the experiment. Reachability at the network layer is evaluated through ICMPv6 probes and, in the last scenario, also through IPv4 tests, in order to verify whether the device responds under the tested perimeter configuration. Service accessibility is evaluated through TCP connection attempts and application layer interactions, including port scanning, HTTP requests, and Telnet sessions. Finally, traffic visibility is evaluated through packet captures and logs generated by the device itself, using tools such as Wireshark to inspect the exchanged traffic and determine whether information such as HTTP requests, server responses, credentials, and commands remains observable in plaintext. For this purpose, standard network analysis tools are used. Reachability tests are performed using `ping6` and `ping`, the state of TCP services is observed with `nmap`, interaction with the HTTP server is validated using `curl`, and Telnet sessions are documented through connection transcripts. At the traffic level, `tcpdump` is used, and when needed, captures are analyzed in Wireshark to inspect packet contents. These external observations are complemented with the ESP32 serial output and the logging mechanisms implemented in the firmware, which record inbound connections, requested resources, authentication attempts, and executed commands. Within this work, a device is considered reachable when it produces valid responses at the network level under the tested configuration. A service is considered accessible when a TCP connection can be established and a valid response is obtained at the application level. In the same way, traffic is considered visible when the exchanged content can be directly inspected in the captures without the need for decryption. These criteria are applied consistently across all scenarios, so that the observed changes can be attributed to the experimental conditions and not to differences in the measurement method.

Table 3.3: Measurement Dimensions and Evidence Sources

Measurement Dimension	Operational Evidence	Primary Tools / Sources
Network reachability	valid ICMP responses under the tested configuration	ping6, ping, serial logs
TCP service state	port state observed as open, filtered, or closed/unreachable	nmap, TCP connection attempts
HTTP accessibility	valid HTTP response received from the root endpoint or hidden endpoint	curl, HTTP output, serial logs
Telnet accessibility	successful session establishment and application-layer interaction	Telnet transcript, serial logs
Traffic visibility	readable application-layer content visible in packet captures without decryption	tcpdump, Wireshark
Device-side logging evidence	locally recorded evidence of inbound connections, endpoint access, and authentication attempts	ESP32 serial output, firmware logs

3.5 Ethical Considerations The experiments are conducted within a controlled infrastructure. The experimental setup is limited to a residential network, an external VPS under the researcher’s control, and an ESP32 device with firmware developed specifically for the purposes of this work. No third-party devices, production systems, or publicly accessible IoT deployments are included in the testbed. The design does not include internet-wide scanning or any interaction with systems outside the experimental environment. Traffic inspection is limited to communications involving the test device and researcher-controlled hosts, keeping the methodology within a closed and ethically controlled setup.

4. Experimental Results

4.1 Address Assignment and Identifier Exposure Upon connecting to a residential WiFi network, the ESP32 successfully completed the IPv6 autoconfiguration process through SLAAC. The logs confirm the assignment of a link-local address as well as a global IPv6 address, together with the MAC address and the parameters observed in Table 4.1.

Table 4.1: Device Identity Parameters

Parameter	Value
MAC address	28:05:A5:0E:73:6C
Link-local IPv6	FE80::2A05:A5FF:FE0E:736C
Global IPv6	2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
Address acquisition method	SLAAC (Stateless Address Autoconfiguration)
IID generation method	EUI-64

The link-local address was assigned first, followed by the global address after the reception of the Router Advertisement. Both addresses coexisted in the interface during all the experimental scenarios. The interface identifier (IID) of the global address follows the EUI-64 format, embedding the device’s MAC address directly into the IPv6 address. This deterministic generation method has been shown to facilitate remote device identification and physical tracking [16].

4.2 Internal Baseline and Local Service Availability (S0) The ESP32 responds correctly to ICMPv6 probes, and both TCP ports (80 and 23) are reachable, with the implemented services behaved as expected.

Table 4.2: S0 Internal Baseline Results

Test	Tool	Result
ICMPv6 reachability	ping6	4/4 replies; initial RTT spike, then stabilized around 7 ms
Port 80 state	nmap -6 -sT	open
Port 23 state	nmap -6 -sT	open
HTTP root (/)	curl	HTTP 200 OK; dashboard returns MAC and IPv6 information
HTTP hidden endpoint (/hidden_admin)	curl	HTTP 200 OK; Diagnostic Mode: ENABLED
Telnet session	ncat	connection established; login successful; status command executed

Accessing the HTTP root endpoint returns a valid response with device information, and the undocumented endpoint /hidden_admin is also accessible without authentication. Furthermore, a Telnet session can be established using the default credentials (admin/admin). All interactions were recorded in the

ESP32 serial logs, which capture source addresses, requested resources, and authentication events (see Appendix B.1 for the raw logs of this scenario). This confirms that both the device and its services were fully operational before introducing any external interaction.

4.3 External IPv6 Reachability with Firewall Enabled (S1) ICMPv6 probes received responses in all attempts (10/10), with an average RTT of 40.8 ms (Table 4.3).

Table 4.3: S1 Results: External IPv6, Firewall Enabled

Test	Tool	Result
ICMPv6 reachability	ping6 -c 10	10/10 replies, 0% loss, avg RTT 40.8 ms
Port 80 state	nmap -6 -sT	filtered (no-response)
Port 23 state	nmap -6 -sT	filtered (no-response)
HTTP root (/)	curl	connection timed out after 10 s
HTTP hidden endpoint (/hidden_admin)	curl	connection timed out after 10 s
Telnet session	ncat	no connection established

However, no transport-layer connectivity was observed. Ports 80 and 23 were reported as filtered, with no TCP responses. HTTP requests timed out, and no Telnet session could be established. The device did not register any incoming TCP connections during this scenario. Packet captures on the VPS confirmed that only ICMPv6 traffic reached the destination, while all TCP SYNs remained unanswered (supported by logs in Appendix B.2).

4.4 External IPv6 Accessibility with Firewall Disabled (S2) ICMPv6 connectivity remained consistent, with 10/10 responses and an average RTT of 40.4 ms (Table 4.4).

Table 4.4: S2 Results: External IPv6, Firewall Disabled

Test	Tool	Result
ICMPv6 reachability	ping6 -c 10	10/10 replies, 0% loss, avg RTT 40.4 ms
Port 80 state	nmap -sT	open (syn-ack)
Port 23 state	nmap -sT	open (syn-ack)
HTTP root (/)	curl	HTTP 200 OK; response includes MAC and IPv6
HTTP hidden endpoint (/hidden_admin)	curl	HTTP 200 OK; Diagnostic Mode: ENABLED
Telnet session	ncat	connection established; login successful
Plaintext exposure	tcpdump -A	GET /hidden_admin and payload visible in capture

Ports 80 and 23 were reported as open, and both services were accessible from the external host. The HTTP root endpoint returned a valid response, the /hidden_admin endpoint was accessible without authentication, and the Telnet session accepted the credentials admin/admin. HTTP responses included device information, and the logs confirmed incoming connections from the external address along with the associated events (see Appendix B.3). Traffic captures revealed that the HTTP exchange, including the request to /hidden_admin and its response, was completely visible in plaintext.

4.5 Local Traffic Visibility and Plaintext Exposure (S3) Traffic captures allowed direct observation of HTTP requests, full server responses, Telnet authentication prompts, and credentials in plaintext (Table 4.5).

Table 4.5: S3 Results: Local Passive Observation

Observation	Tool	Result
HTTP request	Wireshark	GET /hidden_admin visible in packet payload
HTTP response	Wireshark	HTML body visible, including Diagnostic Mode: ENABLED
Telnet banner	Wireshark	Password: prompt visible in ASCII payload
Telnet credential	Wireshark	admin credential visible in plaintext payload
MAC address	Wireshark	28:05:A5:0E:73:6C as source in frame header

The device's MAC address was visible in the Ethernet headers. Both HTTP and Telnet traffic could be inspected directly without the need for decryption.

4.6 IPv4 Baseline Without Port Forwarding (S4) The router responded to ICMP probes, confirming network-level connectivity (Table 4.6).

Table 4.6: S4 Results: IPv4 Baseline

Test	Tool	Result
IPv4 reachability	ping -c 5	5/5 replies from router, avg RTT 35.6 ms
Port 80 state	nmap -sT	filtered (no-response)
Port 23 state	nmap -sT	filtered (no-response)
HTTP root (/)	curl	connection timed out after 10 s

Ports 80 and 23 were reported as filtered, and HTTP requests timed out with no response. No direct inbound access to the device's services was observed under this configuration.

4.7 Cross-Scenario Summary The results show consistent differences between the internal baseline (S0), external network-level response without application access (S1), external service accessibility (S2), local traffic visibility (S3), and the IPv4 scenario without port forwarding (S4).

Table 4.7: Cross-Scenario Comparative Results

Metric	S0 Int.	S1 Ext.	S2 Ext.	S3 Local	S4 IPv4
Global IPv6 assigned	✓	✓	✓	✓	—
EUI-64 MAC derivable	✓	✓	✓	✓	—
ICMP response rate	100%	100%	100%	—	—
Mean RTT	7 ms	40.8 ms	40.4 ms	—	35.6 ms
Port 80 state	open	filtered	open	—	filtered
Port 23 state	open	filtered	open	—	filtered
HTTP root accessible	✓	✗	✓	—	✗
Hidden endpoint accessible	✓	✗	✓	—	✗
Telnet session established	✓	✗	✓	✓	✗
Plaintext payload visible	—	—	✓	✓	—
Credentials visible	—	—	✓	✓	—
Full service access	✓	✗	✓	✓	✗

The comparative findings summarized in Table 4.7 underscore the core vulnerability identified through this experiment. While the device remained protected in the legacy IPv4 environment without manual port forwarding, the transition to a global IPv6 address space enabled direct, external service exposure as soon as the residential firewall was disabled. Furthermore, the systematic leakage of physical identifiers through EUI-64 addresses indicates that device-level privacy is fundamentally compromised in its default configuration. These results provide the empirical basis for the subsequent analysis of risk and mitigation strategies in the following chapters.

5. Discussion

5.1 Interpretation of Results

5.2 Threats to Validity

6. Limitations and Mitigation

6.1 Limitations

6.2 Recommendations for Network Administrators

6.3 Recommendations for Manufacturers

6.4 End-User Best Practices

7. Conclusion

7.1 Future Work

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A. Source Code

This appendix contains the complete source code for the ESP32 firmware used throughout the experiments. The firmware implements the network services (HTTP and Telnet) over IPv6 and manages the device's network identity.

```
#include <Arduino.h>
#include <WiFi.h>
#include "esp_event.h"
#include "esp_netif.h"
#include "lwip/ip6_addr.h"
#include "lwip/sockets.h"
#include "lwip/inet.h"

const char* ssid = "WN-CB5AEO";
const char* password = "hy2pn6gx6i";

static esp_netif_t* sta_netif = nullptr;

esp_ip6_addr_t g_ll_addr;
esp_ip6_addr_t g_global_addr;
bool g_has_ll = false;
bool g_has_global = false;

// =====
// Counters
// =====
uint32_t http_hits_total = 0;
uint32_t hidden_admin_hits = 0;
uint32_t telnet_connections = 0;
uint32_t telnet_login_success = 0;
uint32_t telnet_login_failure = 0;

// =====
// Timestamp helper (ms since boot)
// =====
String ts() {
    return "[" + String(millis()) + "ms]";
}

// =====
// Helpers
// =====
String ipv6ToString(const esp_ip6_addr_t& addr) {
    char buf[40];
    ip6addr_ntoa_r((const ip6_addr_t*)&addr, buf, sizeof(buf));
```

```

    return String(buf);
}

String sockaddrIn6ToString(const sockaddr_in6* addr) {
    char buf[INET6_ADDRSTRLEN];
    inet_ntop(AF_INET6, &(addr->sin6_addr), buf, sizeof(buf));
    return String(buf);
}

void printIPv6(const esp_ip6_addr_t& addr) {
    Serial.println(ipv6ToString(addr));
}

String readLineFromSocket(int clientSock) {
    String line = "";
    char c;
    while (true) {
        int n = recv(clientSock, &c, 1, 0);
        if (n <= 0) break;
        if (c == '\r') continue;
        if (c == '\n') break;
        line += c;
        if (line.length() > 200) break;
    }
    return line;
}

void printCounters() {
    Serial.println("=== COUNTERS ===");
    Serial.println(" http_hits_total: " + String(http_hits_total));
    Serial.println(" hidden_admin_hits: " + String(hidden_admin_hits));
    Serial.println(" telnet_connections: " + String(telnet_connections));
    Serial.println(" telnet_login_success: " + String(telnet_login_success));
    Serial.println(" telnet_login_failure: " + String(telnet_login_failure));
    Serial.println("=====");
}

// =====
// IPv6 Event Handler
// =====
void onGotIPv6(void* arg, esp_event_base_t event_base,
               int32_t event_id, void* event_data) {
    if (event_base == IP_EVENT && event_id == IP_EVENT_GOT_IP6) {
        ip_event_got_ip6_t* event = (ip_event_got_ip6_t*)event_data;
        Serial.print(ts() + " [IPv6] Address acquired: ");
        printIPv6(event->ip6_info.ip);

        if (sta_netif != nullptr) {

```

```

        if (esp_netif_get_ip6_linklocal(sta_netif, &g_ll_addr) == ESP_OK)
            g_has_ll = true;
        if (esp_netif_get_ip6_global(sta_netif, &g_global_addr) == ESP_OK)
            g_has_global = true;
    }
}
}

// =====
// HTTP Server (port 80)
// =====
void httpServerTask(void* pvParameters) {
    int serverSock = socket(AF_INET6, SOCK_STREAM, IPPROTO_TCP);
    if (serverSock < 0) {
        Serial.println(ts() + " [HTTP] Failed to create IPv6 socket");
        vTaskDelete(NULL);
        return;
    }

    int opt = 1;
    setsockopt(serverSock, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    sockaddr_in6 serverAddr;
    memset(&serverAddr, 0, sizeof(serverAddr));
    serverAddr.sin6_family = AF_INET6;
    serverAddr.sin6_addr = in6addr_any;
    serverAddr.sin6_port = htons(80);

    if (bind(serverSock, (struct sockaddr*)&serverAddr, sizeof(serverAddr)) < 0) {
        Serial.println(ts() + " [HTTP] bind() failed");
        close(serverSock);
        vTaskDelete(NULL);
        return;
    }

    if (listen(serverSock, 5) < 0) {
        Serial.println(ts() + " [HTTP] listen() failed");
        close(serverSock);
        vTaskDelete(NULL);
        return;
    }

    Serial.println(ts() + " [HTTP] Listening on port 80 (IPv6)");

    while (true) {
        sockaddr_in6 clientAddr;
        socklen_t clientLen = sizeof(clientAddr);

```

```

int clientSock = accept(serverSock, (struct sockaddr*)&clientAddr, &clientLen);
if (clientSock < 0) { delay(10); continue; }

String remoteIP = sockaddrIn6ToString(&clientAddr);
uint16_t remotePort = ntohs(clientAddr.sin6_port);

char buffer[1024];
int len = recv(clientSock, buffer, sizeof(buffer) - 1, 0);

if (len > 0) {
    buffer[len] = '\0';
    String request = String(buffer);
    int lineEnd = request.indexOf("\r\n");
    String firstLine = (lineEnd > 0) ? request.substring(0, lineEnd) : request;

    String path = "/";
    if (firstLine.startsWith("GET ")) {
        int secondSpace = firstLine.indexOf(' ', 4);
        if (secondSpace > 4) path = firstLine.substring(4, secondSpace);
    }

    http_hits_total++;
    Serial.println(ts() + " [HTTP] src=" + remoteIP
        + " port=" + String(remotePort)
        + " path=" + path);

    String body;
    String status = "HTTP/1.1 200 OK\r\n";

    if (path == "/") {
        body = "<html><body>";
        body += "<h1>ESP32 IoT Device Dashboard</h1>";
        body += "<p>Status: Online</p>";
        body += "<p>Device: ESP32</p>";
        body += "<p>Service: IPv6 test node</p>";
        body += "<p>Firmware: 1.0-demo</p>";
        body += "<p>MAC: " + WiFi.macAddress() + "</p>";
        if (g_has_global) body += "<p>Global IPv6: " + ipv6ToString(g_global_addr) + "</p>";
        if (g_has_ll) body += "<p>Link-local: " + ipv6ToString(g_ll_addr) + "</p>";
        body += "<hr>";
        body += "<p>HTTP hits: " + String(http_hits_total) + "</p>";
        body += "<p>Hidden admin hits: " + String(hidden_admin_hits) + "</p>";
        body += "<p>Telnet connections: " + String(telnet_connections) + "</p>";
        body += "</body></html>";
    }
    else if (path == "/hidden_admin") {
        hidden_admin_hits++;
        Serial.println(ts() + " [HTTP] *** HIDDEN ADMIN accessed from " + remoteIP + " ***");
    }
}

```

```

        body = "<html><body>";
        body += "<h1>Hidden Admin Interface</h1>";
        body += "<p>Diagnostic Mode: ENABLED</p>";
        body += "<p>Debug Access: TRUE</p>";
        body += "<p>Warning: undocumented endpoint</p>";
        body += "<p>MAC: " + WiFi.macAddress() + "</p>";
        body += "</body></html>";
    }
    else {
        status = "HTTP/1.1 404 Not Found\r\n";
        body = "<html><body><h1>404 Not Found</h1></body></html>";
    }

    String response = status
        + "Content-Type: text/html\r\n"
        + "Content-Length: " + String(body.length()) + "\r\n"
        + "Connection: close\r\n\r\n"
        + body;

    send(clientSock, response.c_str(), response.length(), 0);
}

close(clientSock);
printCounters();
}
}

// =====
// Telnet-like TCP Service (port 23)
// =====
void telnetServerTask(void* pvParameters) {
    int serverSock = socket(AF_INET6, SOCK_STREAM, IPPROTO_TCP);
    if (serverSock < 0) {
        Serial.println(ts() + " [TELNET] Failed to create IPv6 socket");
        vTaskDelete(NULL);
        return;
    }

    int opt = 1;
    setsockopt(serverSock, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    sockaddr_in6 serverAddr;
    memset(&serverAddr, 0, sizeof(serverAddr));
    serverAddr.sin6_family = AF_INET6;
    serverAddr.sin6_addr = in6addr_any;
    serverAddr.sin6_port = htons(23);

    if (bind(serverSock, (struct sockaddr*)&serverAddr, sizeof(serverAddr)) < 0) {

```

```

Serial.println(ts() + " [TELNET] bind() failed");
close(serverSock);
vTaskDelete(NULL);
return;
}

if (listen(serverSock, 2) < 0) {
    Serial.println(ts() + " [TELNET] listen() failed");
    close(serverSock);
    vTaskDelete(NULL);
    return;
}

Serial.println(ts() + " [TELNET] Listening on port 23 (IPv6)");

while (true) {
    sockaddr_in6 clientAddr;
    socklen_t clientLen = sizeof(clientAddr);

    int clientSock = accept(serverSock, (struct sockaddr*)&clientAddr, &clientLen);
    if (clientSock < 0) { delay(10); continue; }

    telnet_connections++;
    String remoteIP = sockaddrIn6ToString(&clientAddr);
    uint16_t remotePort = ntohs(clientAddr.sin6_port);

    Serial.println(ts() + " [TELNET] src=" + remoteIP
        + " port=" + String(remotePort) + " connected");

    String banner = "ESP32 Legacy Telnet Service\r\nUsername: ";
    send(clientSock, banner.c_str(), banner.length(), 0);

    String user = readLineFromSocket(clientSock);
    Serial.println(ts() + " [TELNET] src=" + remoteIP + " username=" + user);

    String askPass = "Password: ";
    send(clientSock, askPass.c_str(), askPass.length(), 0);

    String pass = readLineFromSocket(clientSock);
    Serial.println(ts() + " [TELNET] src=" + remoteIP + " password=" + pass);

    if (user == "admin" && pass == "admin") {
        telnet_login_success++;
        Serial.println(ts() + " [TELNET] src=" + remoteIP + " login=SUCCESS");

        String ok = "\r\nAccess granted\r\nType 'status' or 'exit'\r\n> ";
        send(clientSock, ok.c_str(), ok.length(), 0);
    }
}

```

```

while (true) {
    String cmd = readLineFromSocket(clientSock);
    if (cmd.length() == 0) break;

    Serial.println(ts() + " [TELNET] src=" + remoteIP + " cmd=" + cmd);

    if (cmd == "exit") {
        String bye = "Bye\r\n";
        send(clientSock, bye.c_str(), bye.length(), 0);
        break;
    } else if (cmd == "status") {
        String msg = "Device status: ONLINE\r\n";
        msg += "MAC: " + WiFi.macAddress() + "\r\n";
        if (g_has_global) msg += "Global IPv6: " + ipv6ToString(g_global_addr) + "\r\n";
        msg += "HTTP hits: " + String(http_hits_total) + "\r\n";
        msg += "Telnet sessions: " + String(telnet_connections) + "\r\n";
        msg += "Login successes: " + String(telnet_login_success) + "\r\n";
        msg += "> ";
        send(clientSock, msg.c_str(), msg.length(), 0);
    } else {
        String msg = "Unknown command\r\n> ";
        send(clientSock, msg.c_str(), msg.length(), 0);
    }
}

telnet_login_failure++;
Serial.println(ts() + " [TELNET] src=" + remoteIP + " login=FAILED");
String denied = "\r\nAccess denied\r\n";
send(clientSock, denied.c_str(), denied.length(), 0);
}

close(clientSock);
printCounters();
}
}

// =====
// Setup
// =====
void setup() {
    Serial.begin(115200);
    delay(1000);

    WiFi.mode(WIFI_STA);
    WiFi.setSleep(false); // Keep WiFi active prevents disconnection during experiments

    esp_event_handler_register(IP_EVENT, IP_EVENT_GOT_IP6, &onGotIPv6, nullptr);

```

```

Serial.println("=== ESP32 IPv6 Thesis Demo ===");
Serial.println("Connecting to WiFi...");
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Waiting for WiFi...");
}

Serial.println("WiFi connected");
Serial.print("IPv4: "); Serial.println(WiFi.localIP());
Serial.print("MAC: "); Serial.println(WiFi.macAddress());

sta_netif = esp_netif_get_handle_from_ifkey("WIFI_STA_DEF");
if (sta_netif == nullptr) {
    Serial.println("[ERROR] Could not get WIFI_STA_DEF handle");
    return;
}

esp_err_t err = esp_netif_create_ip6_linklocal(sta_netif);
Serial.print("Link-local IPv6 init: ");
Serial.println(err == ESP_OK ? "OK" : "FAILED");

delay(3000); // Wait for SLAAC global address assignment

xTaskCreatePinnedToCore(httpServerTask, "httpTask", 8192, NULL, 1, NULL, 1);
xTaskCreatePinnedToCore(telnetServerTask, "telnetTask", 8192, NULL, 1, NULL, 1);
}

// =====
// Loop prints identity every 100s
// =====
void loop() {
    if (sta_netif != nullptr) {
        esp_ip6_addr_t ll_addr, global_addr;
        esp_err_t ll_ok = esp_netif_get_ip6_linklocal(sta_netif, &ll_addr);
        esp_err_t gl_ok = esp_netif_get_ip6_global(sta_netif, &global_addr);

        Serial.println("=== DEVICE IDENTITY ===");
        Serial.print("MAC: "); Serial.println(WiFi.macAddress());
        Serial.print("IPv4: "); Serial.println(WiFi.localIP());

        if (ll_ok == ESP_OK) {
            g_ll_addr = ll_addr; g_has_ll = true;
            Serial.print("Link-local: "); printIPv6(ll_addr);
        } else {
            Serial.println("Link-local: not yet available");
        }
    }
}

```

```
    if (gl_ok == ESP_OK) {
        g_global_addr = global_addr; g_has_global = true;
        Serial.print("Global IPv6: "); printIPv6(global_addr);
    } else {
        Serial.println("Global IPv6: not yet available");
    }

    Serial.println("=====");
    printCounters();
}

delay(100000);
}
```

Listing A.1: ESP32 Firmware Source Code (main.cpp)

B. Experimental Results

The following sections document the raw evidence collected during the execution of each experimental scenario. This data includes terminal outputs, network probes, and device-level logs.

B.1 Scenario 0: Baseline (Internal Network) The objective of this scenario is to verify the internal availability of all services on the LAN before any external reachability tests.

B.1.1 Local Network Discovery and ICMPv6 The following logs show the device identifying its IPv6 global and link-local addresses.

```
=== ESP32 IPv6 Thesis Demo ===
Connecting to WiFi...
Waiting for WiFi...
WiFi connected
IPv4: 192.168.0.10
MAC: 28:05:A5:0E:73:6C
Link-local IPv6 init: OK
[4028ms] [IPv6] Address acquired: FE80::2A05:A5FF:FE0E:736C
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: not yet available
[5137ms] [TELNET] Listening on port 23 (IPv6)
=====
=== COUNTERS ===
  http_hits_total: 0
  hidden_admin_hits: 0
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
[5137ms] [HTTP] Listening on port 80 (IPv6)
[8028ms] [IPv6] Address acquired: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 0
  hidden_admin_hits: 0
  telnet_connections: 0
```

```

telnet_login_success: 0
telnet_login_failure: 0
=====
                                                                28:05:A5:0E:73:6C

IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 0
  hidden_admin_hits: 0
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
                                                                9AE1:B100:910E:1492:DE64:3BBD port=18403 path=/

=== COUNTERS ===
  http_hits_total: 1
  hidden_admin_hits: 0
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
[274167ms] [HTTP] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=45657 path=/hidden_admin
[274167ms] [HTTP] *** HIDDEN ADMIN accessed from 2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD ***
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
                                                                28:05:A5:0E:73:6C

IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C

```

```

Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 0
    telnet_login_success: 0
    telnet_login_failure: 0
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 0
    telnet_login_success: 0
    telnet_login_failure: 0
=====
[593275ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=4575 connected
[600817ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD username= ' admin
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 1
    telnet_login_success: 0
    telnet_login_failure: 0
=====
[610552ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD password=admin
[610553ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD login=FAILED
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 1
    telnet_login_success: 0
    telnet_login_failure: 1
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C

```

```

IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 1
  telnet_login_success: 0
  telnet_login_failure: 1
=====
[727179ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=17466 connected
[738311ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD username=
[740739ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD password=
[740740ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD login=FAILED
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 2
  telnet_login_success: 0
  telnet_login_failure: 2
=====
[752131ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=17521 connected
[755063ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD username=admin
[757505ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD password=admin
[757505ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD login=SUCCESS
[762498ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD cmd=status
[767211ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD cmd=exit
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 3
  telnet_login_success: 1
  telnet_login_failure: 2
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 3
  telnet_login_success: 1
  telnet_login_failure: 2
=====

```


Listing B.1: S0 - Device Serial Log

B.1.2 Internal Service Verification

```

StatusCode : 200
StatusDescription : OK
Content : <html><body><h1>ESP32 IoT Device Dashboard</h1><p>Status: Online</p><p>Device:
          ESP32</p><p>Service: IPv6 test node</p><p>Firmware: 1.0-demo</p><p>MAC:
          28:05:A5:0E:73:6C</p><p>Global IPv6: 2A02:21B4:9A...
RawContent : HTTP/1.1 200 OK
              Connection: close
              Content-Length: 367
              Content-Type: text/html

              <html><body><h1>ESP32 IoT Device Dashboard</h1><p>Status: Online</p><p>Device:
              ESP32</p><p>Service: IPv6 test node</p>...
Forms : {}
Headers : {[Connection, close], [Content-Length, 367], [Content-Type, text/html]}
Images : {}
InputFields : {}
Links : {}
ParsedHtml : mshtml.HTMLDocumentClass
RawContentLength : 367

```

Listing B.2: S0 - Local HTTP Root Request

```

StatusCode : 200
StatusDescription : OK
Content : <html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access:
          TRUE</p><p>Warning: undocumented endpoint</p><p>MAC: 28:05:A5:0E:73:6C</p></body></ht
RawContent : HTTP/1.1 200 OK
              Connection: close
              Content-Length: 179
              Content-Type: text/html

              <html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Ac
              TRUE</p><p>Warning: undoc...
Forms : {}
Headers : {[Connection, close], [Content-Length, 179], [Content-Type, text/html]}
Images : {}
InputFields : {}
Links : {}
ParsedHtml : mshtml.HTMLDocumentClass
RawContentLength : 179

```

Listing B.3: S0 - Local HTTP Hidden Admin Request

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 16:53 +0100
Nmap scan report for esp32-0e736c.home (2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c)
Host is up (0.0084s latency).

PORT STATE SERVICE
23/tcp open telnet
80/tcp open http

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

Listing B.4: S0 - Local Nmap Scan Output

B.2 Scenario 1: External Reachability (Firewall ON) In this scenario, an external VPS attempts to reach the device while the residential router's IPv6 firewall is active.

B.2.1 Remote Reachability and Service Probing

```
PING 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C (2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c) 56 data bytes
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=1 ttl=238 time=41.9 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=2 ttl=238 time=44.1 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=3 ttl=238 time=38.6 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=4 ttl=238 time=39.1 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=5 ttl=238 time=38.8 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=6 ttl=238 time=41.7 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=7 ttl=238 time=38.7 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=8 ttl=238 time=43.8 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=9 ttl=238 time=41.0 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=10 ttl=238 time=40.1 ms

--- 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 38.627/40.787/44.108/1.964 ms
```

Listing B.5: S1 - External ICMPv6 Ping

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-25 16:12 UTC
Nmap scan report for 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Host is up, received user-set.

PORT STATE SERVICE REASON
23/tcp filtered telnet no-response
80/tcp filtered http no-response

Nmap done: 1 IP address (1 host up) scanned in 3.46 seconds
```

Listing B.6: S1 - External Nmap Scan (Firewall Active)

B.3 Scenario 2: Service Exposure (Firewall OFF) This scenario demonstrates the full exposure of the IoT services when the IPv6 firewall is disabled or misconfigured.

B.3.1 External Service Access

```
ssh : % Total % Received % Xferd Average Speed Time Time Time
      Current
At line:1 char:1
+ ssh $VPS "curl -v --max-time 10 http://[$ESP32]/hidden_admin" 2>&1 ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: ( % Total % ... Time Curre
nt:String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

Dload Upload Total Spent Left Speed

 0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:--
0* Trying [2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c]:80...
* Connected to 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
(2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c) port 80
> GET /hidden_admin HTTP/1.1
> Host: [2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: text/html
< Content-Length: 179
< Connection: close
<
{ [179 bytes data]

100 179 100 179 0 0 1757 0 --:--:-- --:--:-- --:--:--
1772
* Closing connection
<html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access: TRUE</p><p>W
```

Listing B.7: S2 - External HTTP Hidden Admin Access

```
Trying 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c...
Connected to 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C.
Escape character is '^]'.
ESP32 Legacy Telnet Service
Username: Password:
Access granted
Type 'status' or 'exit'
> Device status: ONLINE
MAC: 28:05:A5:0E:73:6C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
HTTP hits: 2
```

```
Telnet sessions: 2
Login successes: 1
> Bye
```

Listing B.8: S2 - External Telnet Session Attempt

B.3.2 Packet Analysis and Remote Discovery

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-25 16:18 UTC
Nmap scan report for 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Host is up, received user-set (0.045s latency).

PORT STATE SERVICE REASON
23/tcp open  telnet syn-ack
80/tcp open  http  syn-ack

Nmap done: 1 IP address (1 host up) scanned in 0.26 seconds
```

Listing B.9: S2 - Remote Port Scan Result

B.4 Scenario 3: Local Traffic Observation This scenario validates that sensitive information remains exposed in plaintext to local observers within the LAN.

B.4.1 Deep Packet Inspection (DPI) The following captures demonstrate that HTTP and Telnet payloads are visible within the network segment.

```
No. Time Source Destination Protocol Length Info
 62433 123.312874 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
 86 6250 -> 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM

Frame 62433: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{BE969710-4
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:14.458268000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:14.458268000 UTC
  Epoch Arrival Time: 1774456274.458268000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 6.246000 milliseconds]
  [Time since reference or first frame: 2 minutes, 3.312874000 seconds]
  Frame Number: 62433
  Frame Length: 86 bytes (688 bits)
  Capture Length: 86 bytes (688 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http tcp.port == 80 http2]
```

Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
 Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
 Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 0, Len: 0

No. Time Source Destination Protocol Length Info

62586 123.634538 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP
 78 80 -> 6250 [SYN, ACK] Seq=0 Ack=1 Win=5760 Len=0 MSS=1412

Frame 62586: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:14.779932000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:14.779932000 UTC

Epoch Arrival Time: 1774456274.779932000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 7.760000 milliseconds]

[Time delta from previous displayed frame: 321.664000 milliseconds]

[Time since reference or first frame: 2 minutes, 3.634538000 seconds]

Frame Number: 62586

Frame Length: 78 bytes (624 bits)

Capture Length: 78 bytes (624 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp]

Character encoding: ASCII (0)

[Coloring Rule Name: HTTP]

[Coloring Rule String: http tcp.port == 80 http2]

Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)

Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd

Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 0, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info

62587 123.634614 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
 74 6250 -> 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0

Frame 62587: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:14.780008000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:14.780008000 UTC

Epoch Arrival Time: 1774456274.780008000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 76.000 microseconds]

[Time delta from previous displayed frame: 76.000 microseconds]

[Time since reference or first frame: 2 minutes, 3.634614000 seconds]

Frame Number: 62587

```

Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info
62588 123.635123 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c HTTP
272 GET /hidden_admin HTTP/1.1

Frame 62588: Packet, 272 bytes on wire (2176 bits), 272 bytes captured (2176 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.780517000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.780517000 UTC
Epoch Arrival Time: 1774456274.780517000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 509.000 microseconds]
[Time delta from previous displayed frame: 509.000 microseconds]
[Time since reference or first frame: 2 minutes, 3.635123000 seconds]
Frame Number: 62588
Frame Length: 272 bytes (2176 bits)
Capture Length: 272 bytes (2176 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:http]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 1, Ack: 1, Len: 198
Hypertext Transfer Protocol
GET /hidden_admin HTTP/1.1\r\n
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) WindowsPowerShell/5.1.26100.7920\r\n
Host: [2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]\r\n
Connection: Keep-Alive\r\n
\r\n
[Response in frame: 62595]
[Full request URI: http://[2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]/hidden_admin]

```

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No. Time Source Destination Protocol Length Info
62595 123.656565 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd HTTP
337 HTTP/1.1 200 OK (text/html)

Frame 62595: Packet, 337 bytes on wire (2696 bits), 337 bytes captured (2696 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:14.801959000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:14.801959000 UTC
  Epoch Arrival Time: 1774456274.801959000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 16.340000 milliseconds]
  [Time delta from previous displayed frame: 21.442000 milliseconds]
  [Time since reference or first frame: 2 minutes, 3.656565000 seconds]
  Frame Number: 62595
  Frame Length: 337 bytes (2696 bits)
  Capture Length: 337 bytes (2696 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:http:data-text-lines]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 1, Ack: 199, Len: 263
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
  Content-Type: text/html\r\n
  Content-Length: 179\r\n
  Connection: close\r\n
  \r\n
  [Request in frame: 62588]
  [Time since request: 21.442000 milliseconds]
  [Request URI: /hidden_admin]
  [Full request URI: http://[2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]/hidden_admin]
  File Data: 179 bytes
Line-based text data: text/html (1 lines)
  <html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access: TRUE</p>

No. Time Source Destination Protocol Length Info
62597 123.660383 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP
74 80 -> 6250 [FIN, ACK] Seq=264 Ack=199 Win=5562 Len=0

Frame 62597: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
```

```

Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.805777000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.805777000 UTC
Epoch Arrival Time: 1774456274.805777000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 94.000 microseconds]
[Time delta from previous displayed frame: 3.818000 milliseconds]
[Time since reference or first frame: 2 minutes, 3.660383000 seconds]
Frame Number: 62597
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 264, Ack: 199, Len: 0

No. Time Source Destination Protocol Length Info
62598 123.660416 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6250 -> 80 [ACK] Seq=199 Ack=265 Win=65272 Len=0

Frame 62598: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.805810000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.805810000 UTC
Epoch Arrival Time: 1774456274.805810000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 33.000 microseconds]
[Time delta from previous displayed frame: 33.000 microseconds]
[Time since reference or first frame: 2 minutes, 3.660416000 seconds]
Frame Number: 62598
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 199, Ack: 265, Len: 0

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No. Time Source Destination Protocol Length Info

64162 126.585140 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6250 -> 80 [FIN, ACK] Seq=199 Ack=265 Win=65272 Len=0

Frame 64162: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:17.730534000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:17.730534000 UTC

Epoch Arrival Time: 1774456277.730534000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 4.733000 milliseconds]

[Time delta from previous displayed frame: 2.924724000 seconds]

[Time since reference or first frame: 2 minutes, 6.585140000 seconds]

Frame Number: 64162

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp]

Character encoding: ASCII (0)

[Coloring Rule Name: HTTP]

[Coloring Rule String: http tcp.port == 80 http2]

Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)

Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c

Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 199, Ack: 265, Len: 0

No. Time Source Destination Protocol Length Info

64163 126.593359 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP
74 80 -> 6250 [ACK] Seq=265 Ack=200 Win=5561 Len=0

Frame 64163: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:17.738753000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:17.738753000 UTC

Epoch Arrival Time: 1774456277.738753000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 8.219000 milliseconds]

[Time delta from previous displayed frame: 8.219000 milliseconds]

[Time since reference or first frame: 2 minutes, 6.593359000 seconds]

Frame Number: 64163

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

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[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 265, Ack: 200, Len: 0

No. Time Source Destination Protocol Length Info
 85498 166.512013 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
86 6336 -> 23 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM

Frame 85498: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{BE969710-4
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:57.657407000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:57.657407000 UTC
  Epoch Arrival Time: 1774456317.657407000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 2.383000 milliseconds]
  [Time delta from previous displayed frame: 39.918654000 seconds]
  [Time since reference or first frame: 2 minutes, 46.512013000 seconds]
  Frame Number: 85498
  Frame Length: 86 bytes (688 bits)
  Capture Length: 86 bytes (688 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP SYN/FIN]
  [Coloring Rule String: tcp.flags & 0x02 tcp.flags.fin == 1]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 0, Len: 0

No. Time Source Destination Protocol Length Info
 85499 166.520588 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP
78 23 -> 6336 [SYN, ACK] Seq=0 Ack=1 Win=5760 Len=0 MSS=1412

Frame 85499: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{BE969710-4
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:57.665982000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:57.665982000 UTC
  Epoch Arrival Time: 1774456317.665982000

```

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[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 8.575000 milliseconds]
[Time delta from previous displayed frame: 8.575000 milliseconds]
[Time since reference or first frame: 2 minutes, 46.520588000 seconds]
Frame Number: 85499
Frame Length: 78 bytes (624 bits)
Capture Length: 78 bytes (624 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP SYN/FIN]
[Coloring Rule String: tcp.flags & 0x02 tcp.flags.fin == 1]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 0, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info
85500 166.520657 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=1 Ack=1 Win=65535 Len=0

Frame 85500: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:57.666051000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:57.666051000 UTC
Epoch Arrival Time: 1774456317.666051000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 69.000 microseconds]
[Time delta from previous displayed frame: 69.000 microseconds]
[Time since reference or first frame: 2 minutes, 46.520657000 seconds]
Frame Number: 85500
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 1, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info
85503 166.530566 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET
113 39 bytes data

```

```

Frame 85503: Packet, 113 bytes on wire (904 bits), 113 bytes captured (904 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:57.675960000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:57.675960000 UTC
  Epoch Arrival Time: 1774456317.675960000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 5.904000 milliseconds]
  [Time delta from previous displayed frame: 9.909000 milliseconds]
  [Time since reference or first frame: 2 minutes, 46.530566000 seconds]
  Frame Number: 85503
  Frame Length: 113 bytes (904 bits)
  Capture Length: 113 bytes (904 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 1, Ack: 1, Len: 39
Telnet

No. Time Source Destination Protocol Length Info
    85536 166.580399 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=1 Ack=40 Win=65496 Len=0

Frame 85536: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:57.725793000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:57.725793000 UTC
  Epoch Arrival Time: 1774456317.725793000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 182.000 microseconds]
  [Time delta from previous displayed frame: 49.833000 milliseconds]
  [Time since reference or first frame: 2 minutes, 46.580399000 seconds]
  Frame Number: 85536
  Frame Length: 74 bytes (592 bits)
  Capture Length: 74 bytes (592 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp]
  Character encoding: ASCII (0)

```

[Coloring Rule Name: TCP] [Coloring Rule String: tcp] Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c) Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 1, Ack: 40, Len: 0	
No. Time Source Destination Protocol Length Info 86964 169.351640 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TELNET 81 7 bytes data	
Frame 86964: Packet, 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8} Section number: 1 Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}) Encapsulation type: Ethernet (1) Arrival Time: Mar 25, 2026 17:32:00.497034000 W. Europe Standard Time UTC Arrival Time: Mar 25, 2026 16:32:00.497034000 UTC Epoch Arrival Time: 1774456320.497034000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 6.060000 milliseconds] [Time delta from previous displayed frame: 2.771241000 seconds] [Time since reference or first frame: 2 minutes, 49.351640000 seconds] Frame Number: 86964 Frame Length: 81 bytes (648 bits) Capture Length: 81 bytes (648 bits) [Frame is marked: False] [Frame is ignored: False] [Protocols in frame: eth:ethertype:ipv6:tcp:telnet] Character encoding: ASCII (0) [Coloring Rule Name: TCP] [Coloring Rule String: tcp]	
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c) Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 1, Ack: 40, Len: 7 Telnet	
No. Time Source Destination Protocol Length Info 86966 169.361376 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET 84 10 bytes data	
Frame 86966: Packet, 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8} Section number: 1 Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}) Encapsulation type: Ethernet (1) Arrival Time: Mar 25, 2026 17:32:00.506770000 W. Europe Standard Time UTC Arrival Time: Mar 25, 2026 16:32:00.506770000 UTC Epoch Arrival Time: 1774456320.506770000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 1.021000 milliseconds]	

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[Time delta from previous displayed frame: 9.736000 milliseconds]
[Time since reference or first frame: 2 minutes, 49.361376000 seconds]
Frame Number: 86966
Frame Length: 84 bytes (672 bits)
Capture Length: 84 bytes (672 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 40, Ack: 8, Len: 10
Telnet

No. Time Source Destination Protocol Length Info
  86992 169.411843 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
  74 6336 -> 23 [ACK] Seq=8 Ack=50 Win=65486 Len=0

Frame 86992: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:32:00.557237000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:32:00.557237000 UTC
Epoch Arrival Time: 1774456320.557237000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 9.385000 milliseconds]
[Time delta from previous displayed frame: 50.467000 milliseconds]
[Time since reference or first frame: 2 minutes, 49.411843000 seconds]
Frame Number: 86992
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 8, Ack: 50, Len: 0

No. Time Source Destination Protocol Length Info
  87954 171.420833 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TELNET
  81 7 bytes data

```

```

Frame 87954: Packet, 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:02.566227000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:02.566227000 UTC
  Epoch Arrival Time: 1774456322.566227000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 741.000 microseconds]
  [Time delta from previous displayed frame: 2.008990000 seconds]
  [Time since reference or first frame: 2 minutes, 51.420833000 seconds]
  Frame Number: 87954
  Frame Length: 81 bytes (648 bits)
  Capture Length: 81 bytes (648 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
  Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
  Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
  Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 8, Ack: 50, Len: 7
  Telnet

No. Time Source Destination Protocol Length Info
    87962 171.439640 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET 119 45 bytes data

Frame 87962: Packet, 119 bytes on wire (952 bits), 119 bytes captured (952 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:02.585034000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:02.585034000 UTC
  Epoch Arrival Time: 1774456322.585034000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 3.458000 milliseconds]
  [Time delta from previous displayed frame: 18.807000 milliseconds]
  [Time since reference or first frame: 2 minutes, 51.439640000 seconds]
  Frame Number: 87962
  Frame Length: 119 bytes (952 bits)
  Capture Length: 119 bytes (952 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]

```

```

[Coloring Rule String: tcp]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 50, Ack: 15, Len: 45
Telnet

No. Time Source Destination Protocol Length Info
  87977 171.482457 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
  74 6336 -> 23 [ACK] Seq=15 Ack=95 Win=65441 Len=0

Frame 87977: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:02.627851000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:02.627851000 UTC
  Epoch Arrival Time: 1774456322.627851000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 2.080000 milliseconds]
  [Time delta from previous displayed frame: 42.817000 milliseconds]
  [Time since reference or first frame: 2 minutes, 51.482457000 seconds]
  Frame Number: 87977
  Frame Length: 74 bytes (592 bits)
  Capture Length: 74 bytes (592 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 15, Ack: 95, Len: 0

No. Time Source Destination Protocol Length Info
  92324 179.380451 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TELNET
  82 8 bytes data

Frame 92324: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:10.525845000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:10.525845000 UTC
  Epoch Arrival Time: 1774456330.525845000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 183.000 microseconds]
  [Time delta from previous displayed frame: 7.897994000 seconds]

```



```

[Time since reference or first frame: 2 minutes, 59.380451000 seconds]
Frame Number: 92324
Frame Length: 82 bytes (656 bits)
Capture Length: 82 bytes (656 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 15, Ack: 95, Len: 8
Telnet

No. Time Source Destination Protocol Length Info
  92325 179.391380 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET 8
  231 157 bytes data

Frame 92325: Packet, 231 bytes on wire (1848 bits), 231 bytes captured (1848 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:10.536774000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:10.536774000 UTC
  Epoch Arrival Time: 1774456330.536774000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 10.929000 milliseconds]
  [Time delta from previous displayed frame: 10.929000 milliseconds]
  [Time since reference or first frame: 2 minutes, 59.391380000 seconds]
  Frame Number: 92325
  Frame Length: 231 bytes (1848 bits)
  Capture Length: 231 bytes (1848 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
  Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
  Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
  Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 95, Ack: 23, Len: 157
  Telnet

No. Time Source Destination Protocol Length Info
  92344 179.444809 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP 74
  6336 -> 23 [ACK] Seq=23 Ack=252 Win=65284 Len=0

```

```

Frame 92344: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:10.590203000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:10.590203000 UTC
  Epoch Arrival Time: 1774456330.590203000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 4.653000 milliseconds]
  [Time delta from previous displayed frame: 53.429000 milliseconds]
  [Time since reference or first frame: 2 minutes, 59.444809000 seconds]
  Frame Number: 92344
  Frame Length: 74 bytes (592 bits)
  Capture Length: 74 bytes (592 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 23, Ack: 252, Len: 0

No. Time Source Destination Protocol Length Info
163041 299.395889 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [RST, ACK] Seq=23 Ack=252 Win=0 Len=0

Frame 163041: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:34:10.541283000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:34:10.541283000 UTC
  Epoch Arrival Time: 1774456450.541283000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 12.566000 milliseconds]
  [Time delta from previous displayed frame: 1 minute, 59.951080000 seconds]
  [Time since reference or first frame: 4 minutes, 59.395889000 seconds]
  Frame Number: 163041
  Frame Length: 74 bytes (592 bits)
  Capture Length: 74 bytes (592 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP RST]
  [Coloring Rule String: tcp.flags.reset eq 1]

```

Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 23, Ack: 252, Len: 0

Listing B.10: S3 - Wireshark Terminal Export (Plaintext Traffic)

B.5 Scenario 4: Comparative Baseline (IPv4 / NAT) The final scenario provides a comparison with a traditional IPv4/NAT environment, where external reachability is denied by default.

B.5.1 IPv4 External Reachability Attempt

```
PING 144.2.66.107 (144.2.66.107) 56(84) bytes of data.
64 bytes from 144.2.66.107: icmp_seq=1 ttl=49 time=39.1 ms
64 bytes from 144.2.66.107: icmp_seq=2 ttl=49 time=34.8 ms
64 bytes from 144.2.66.107: icmp_seq=3 ttl=49 time=34.8 ms
64 bytes from 144.2.66.107: icmp_seq=4 ttl=49 time=34.7 ms
64 bytes from 144.2.66.107: icmp_seq=5 ttl=49 time=34.8 ms

--- 144.2.66.107 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 34.651/35.621/39.055/1.717 ms
```

Listing B.11: S4 - External IPv4 Ping

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-25 16:48 UTC
Nmap scan report for bbs-66-107.pub.wingo.ch (144.2.66.107)
Host is up, received user-set.

PORT STATE SERVICE REASON
23/tcp filtered telnet no-response
80/tcp filtered http no-response

Nmap done: 1 IP address (1 host up) scanned in 3.11 seconds
```

Listing B.12: S4 - External IPv4 Nmap Scan (NAT Filtered)

